

SYSTEM ENGINEERING PART 2

Cyber-Physical Systems 2

Prof. Dr.-Ing. Michael Wiehl



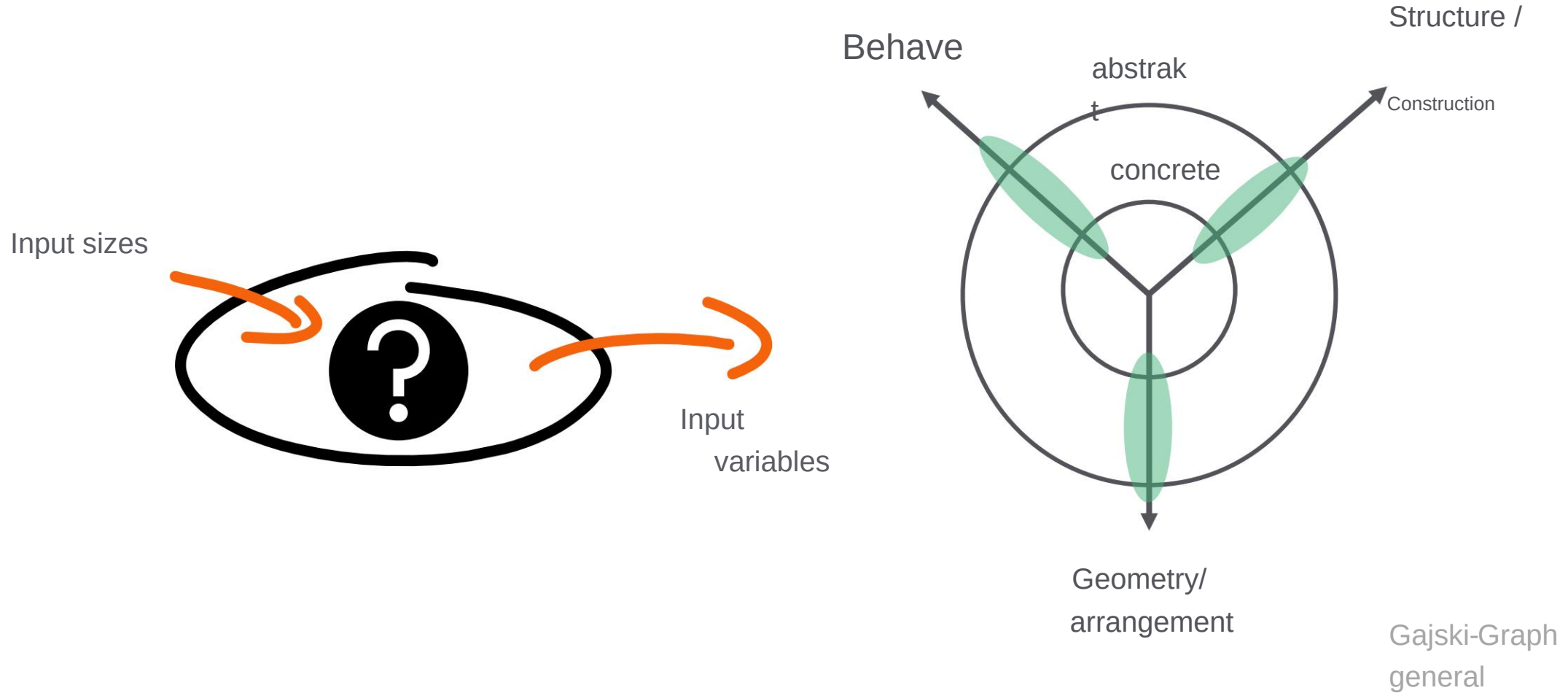
Learning objectives

- Be able to describe systems
- Plan and implement system architecture can
- Be able to choose diagrams for architecture
- Be able to create diagrams in draft status

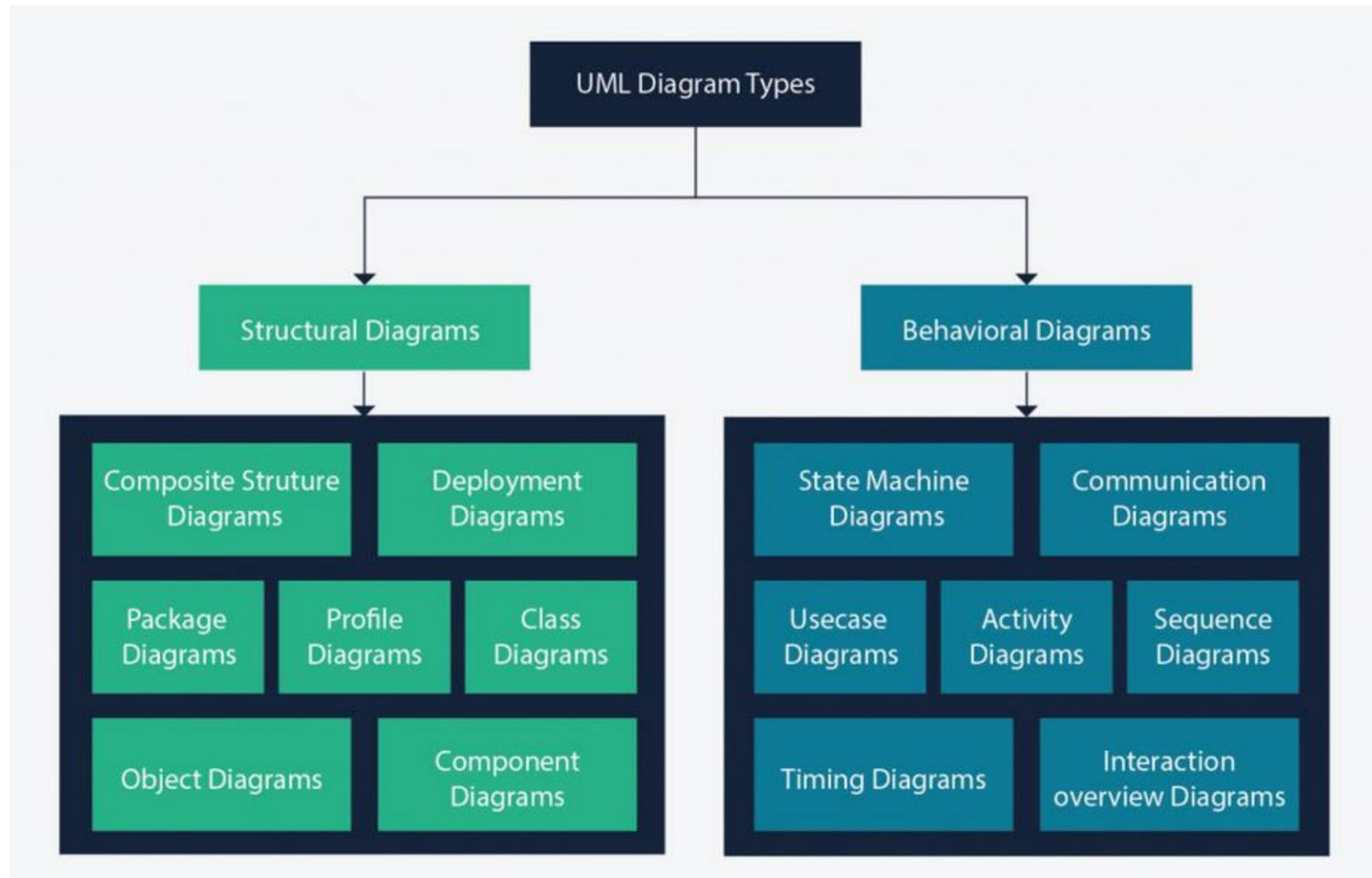


DESCRIPTION- METHODS

How can a system be described?



Software engineering diagrams according to UML

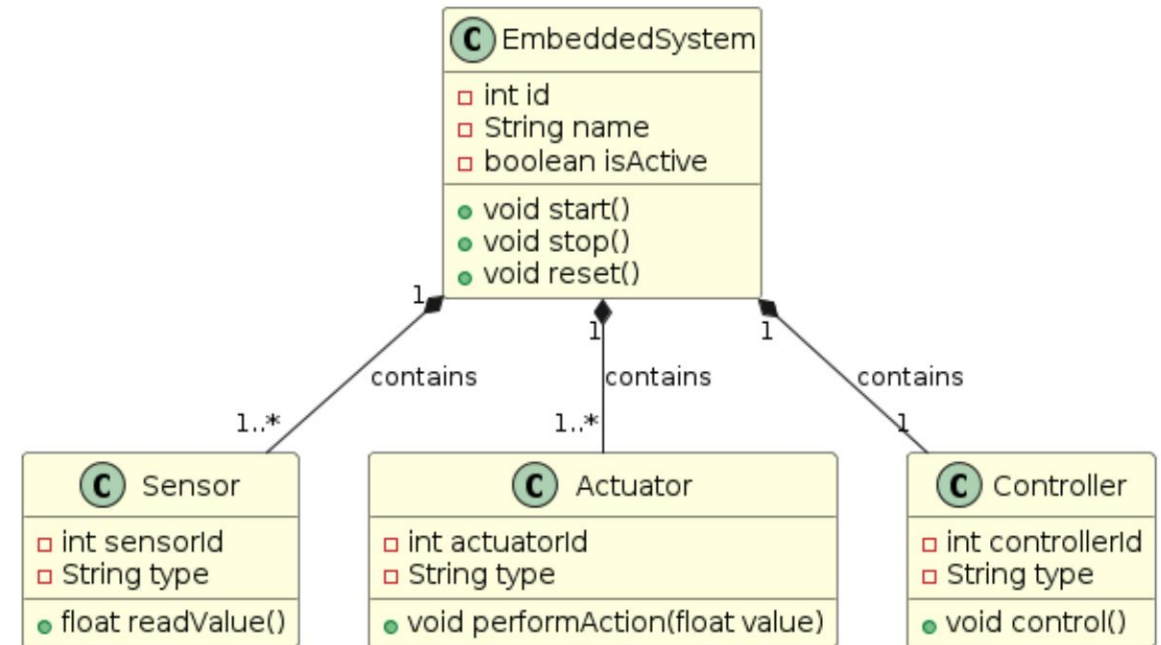
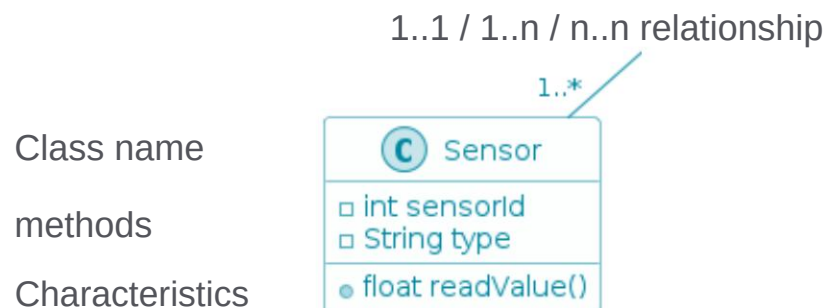


(Source: creately.com)

DESCRIPTION OF SYSTEM STRUCTURE

Class diagram (standardized in UML)

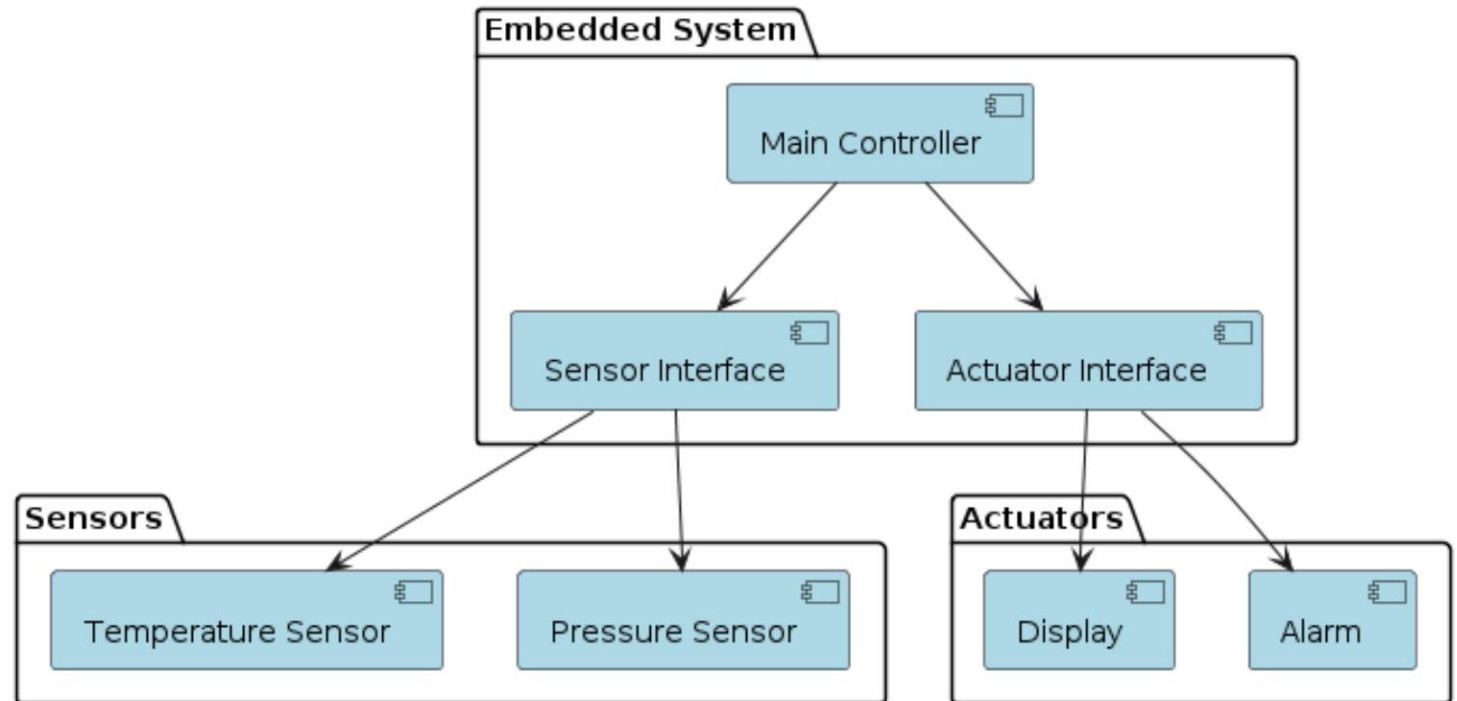
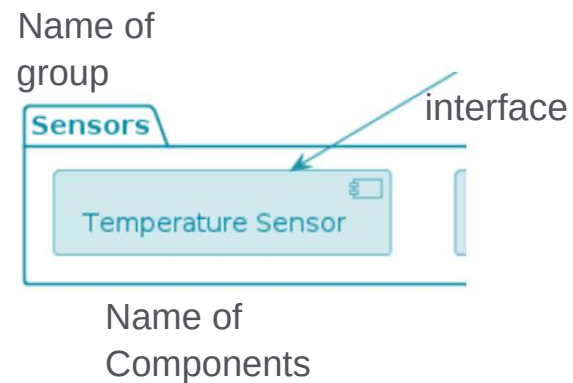
- Graphical modeling of classes, Interfaces and their relationship to each other
- This comprehensively describes the "blueprint" for a data structure



Class diagram for an embedded system in PlantUML

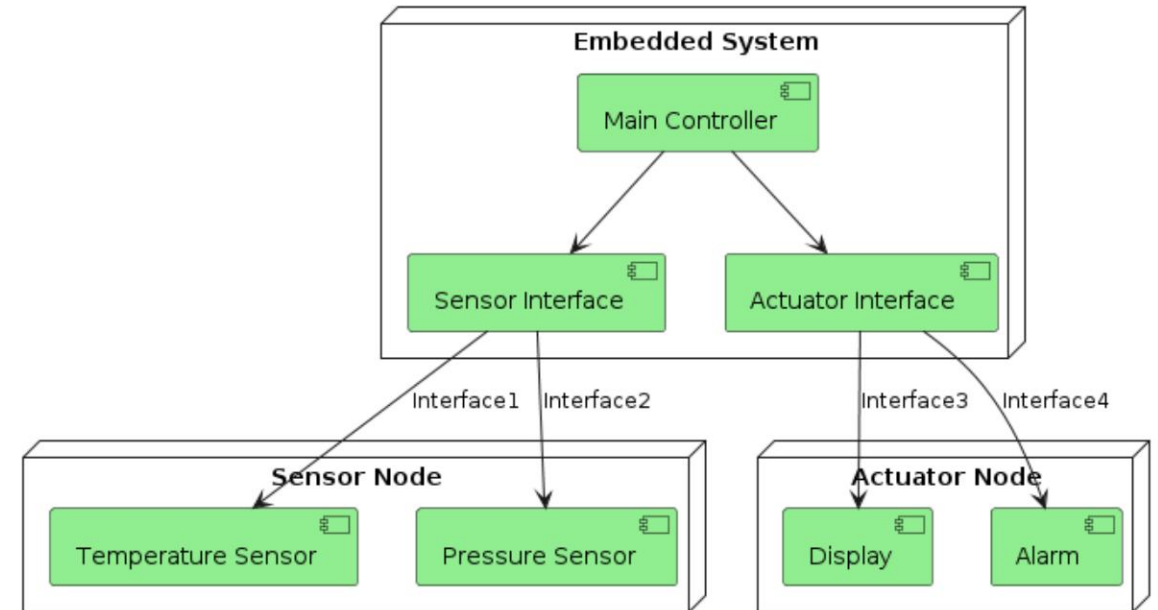
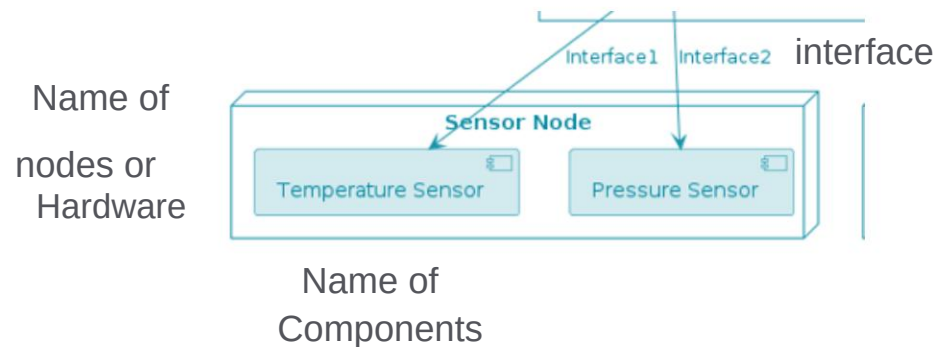
Component diagram (standardized in UML2.3)

- Represents the relationships between the individual components
- This enables the display of System components



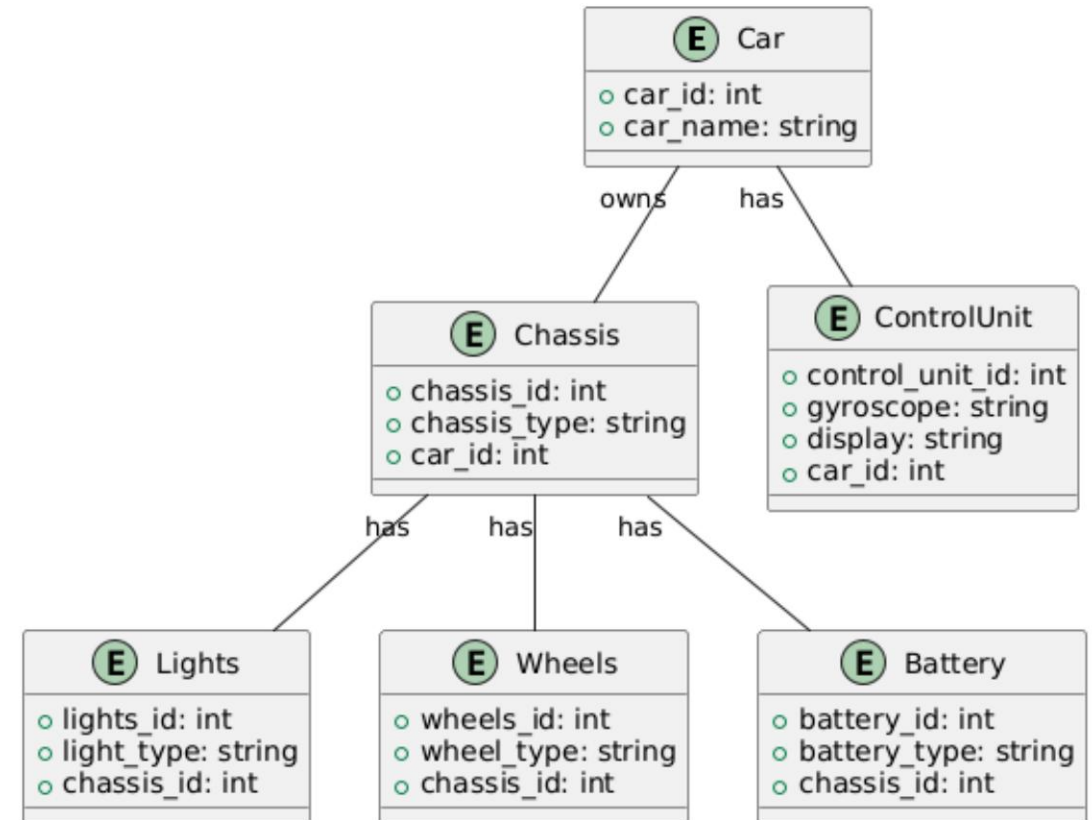
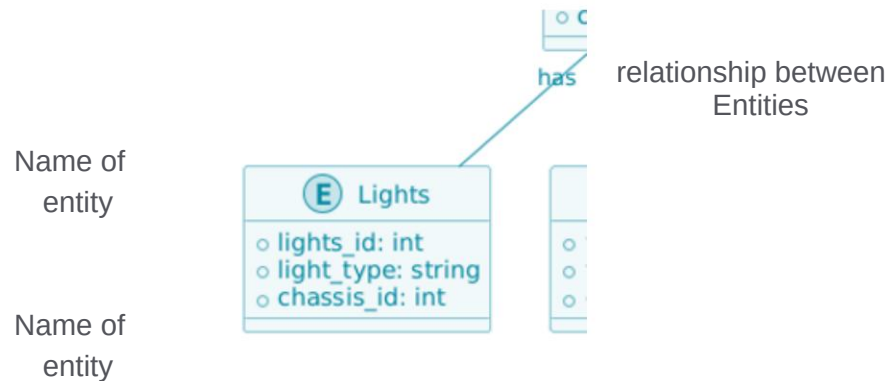
Distribution diagram (standardized in UML2.3)

- Distribution diagrams describe which Components (objects) on which nodes or different hardware run
- Particularly relevant for distributed systems



Entity diagram

- describes relationships between entities such as data
- Used, for example, for database design



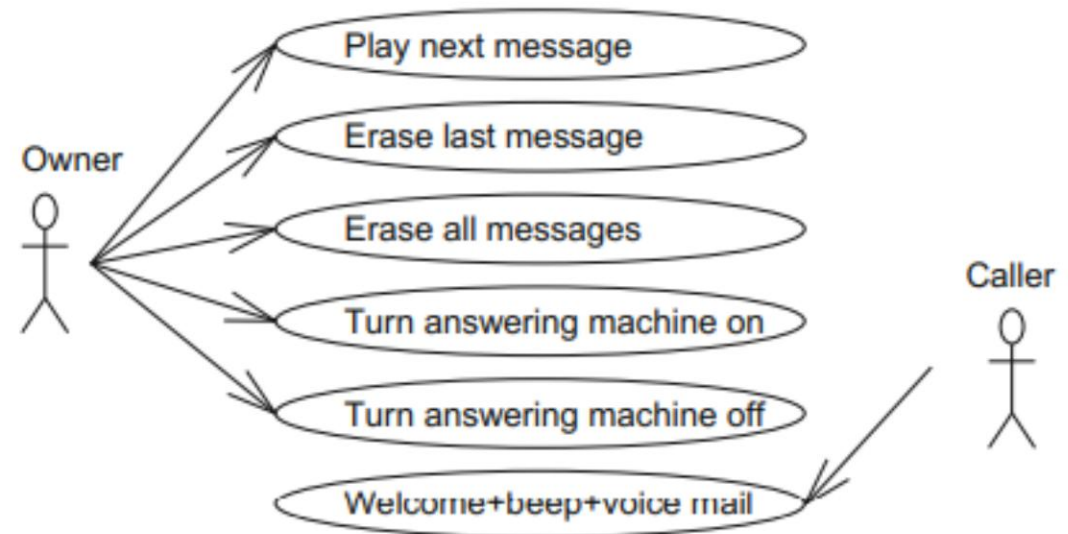
DESCRIPTION OF SYSTEM BEHAVIOR

Use case diagram

Use case diagram

- Quite possible in the early stages
- Use cases can also be presented as prose text be described
- Describes behavior from the user's perspective
- Good for coordination with users/operators
- Helpful for planning development with SCRUM (Keyword: User stories)

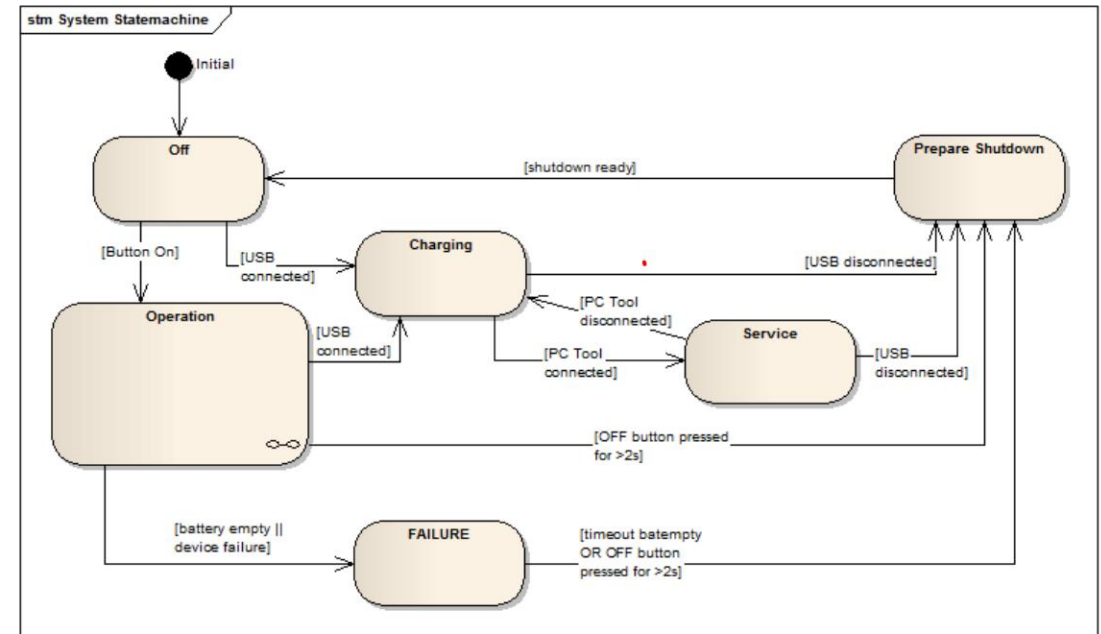
Use cases in a
Answering machine:



State diagram

State (machine) diagram

- The state diagram describes how the
The device behaves externally
- Conditions must be agreed upon by many people in
The design process needs to be understood
- Description for non-experts are
Recommended, for example, to distinguish between
"Off" and "Charging" or "Power down".
explain

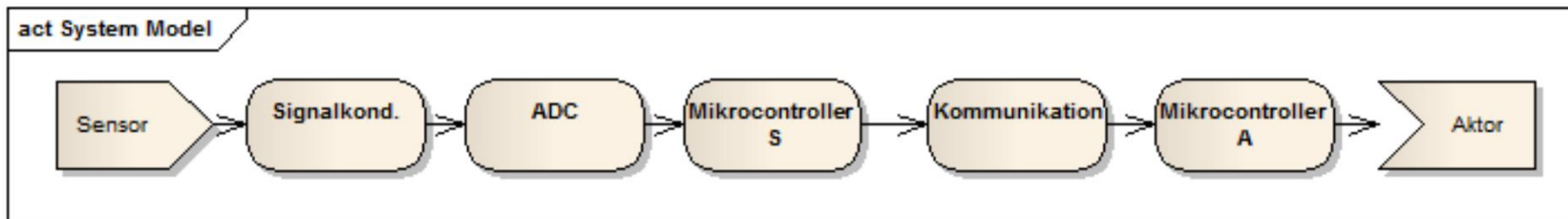


(Note: State machine refers to the
(Implementation, not the diagram)

Chain of effects or signal flow diagrams

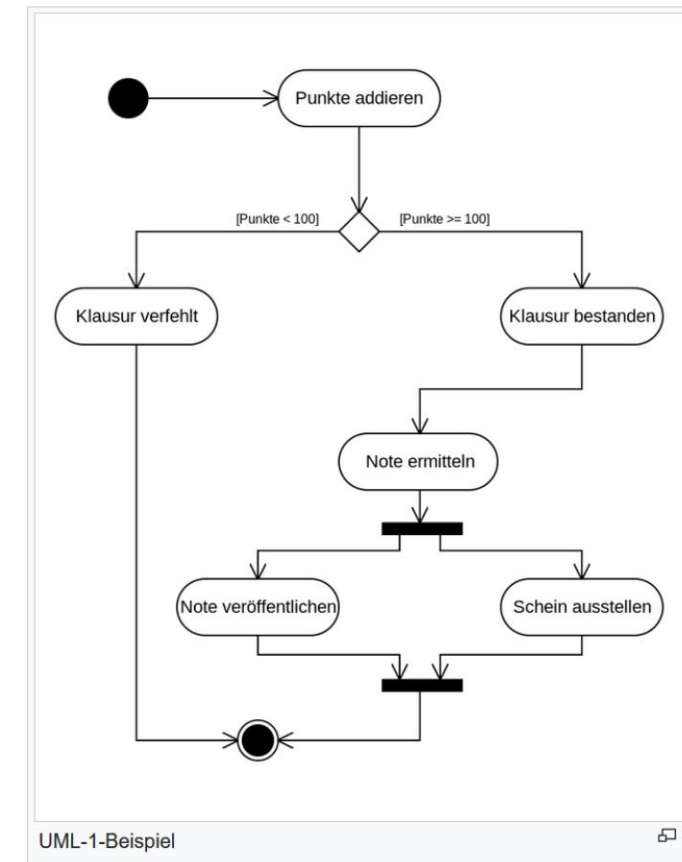
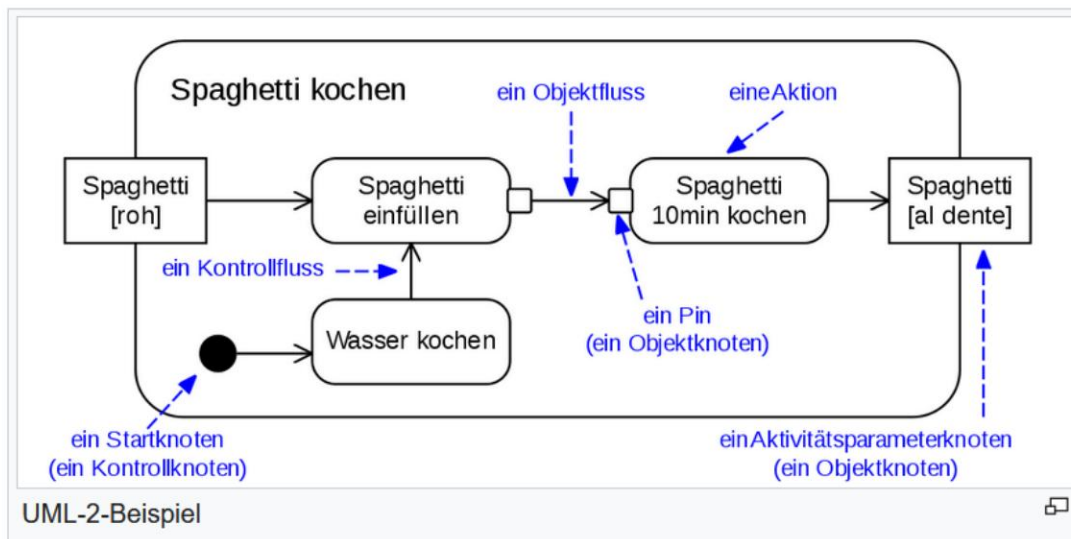
Data flow model

Tracking the signal flow through a system makes sense. It is useful to model flows or components, for example using the UML Data Flow Model.



Activity diagram

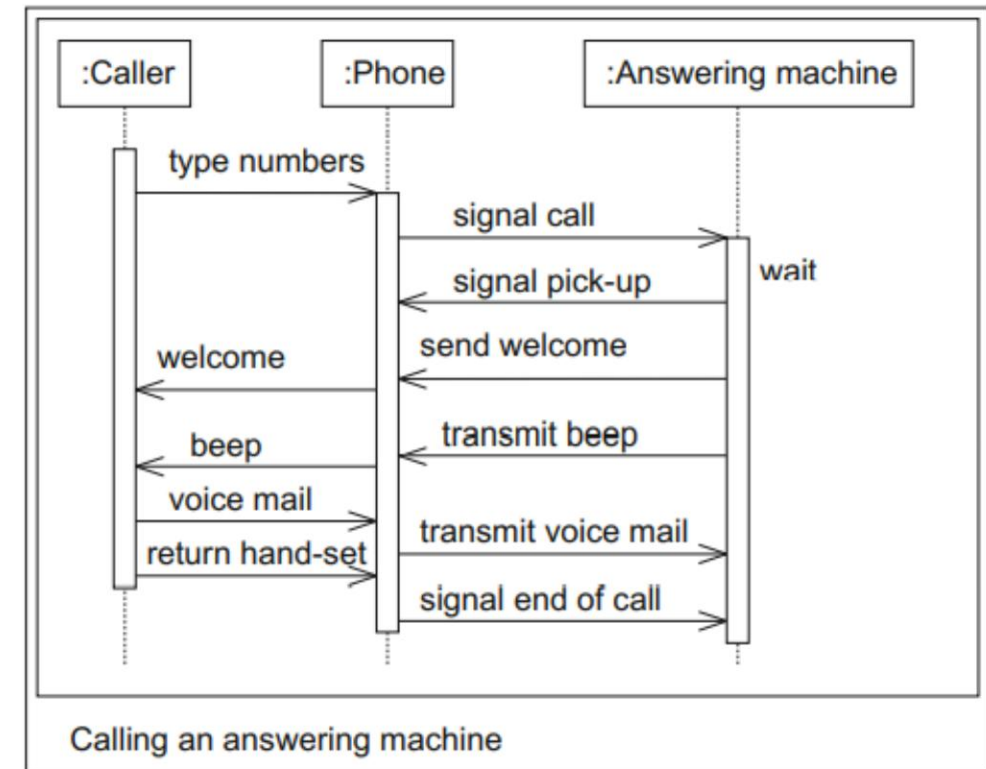
- Visualizes processes or algorithms
- Based on program flowchart



Sequence diagram

sequence diagram

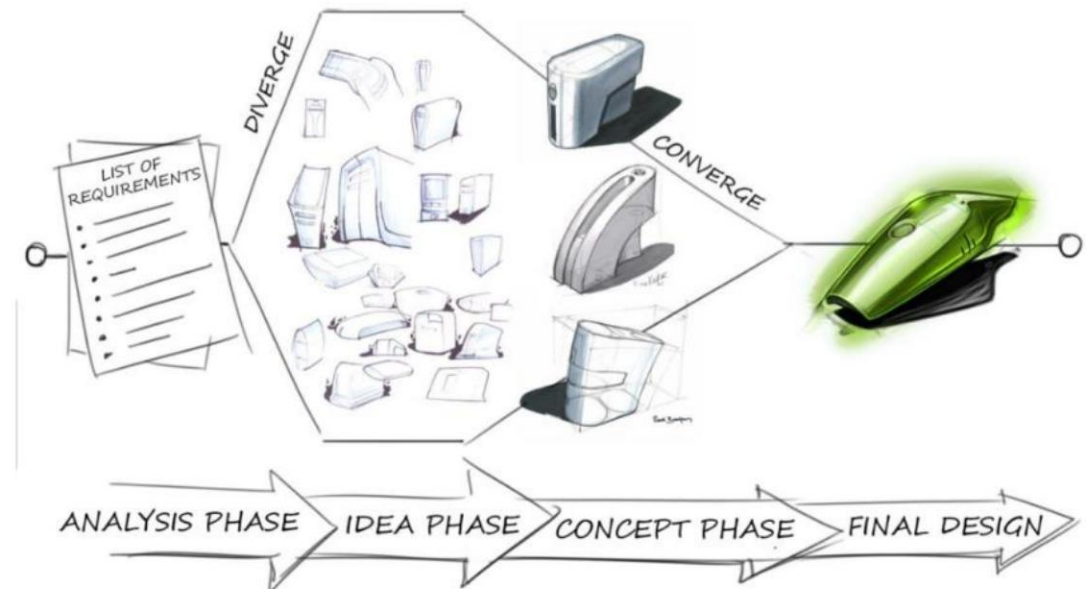
- Describes the process and calls of methods
- Clearly presents processes
- can be affected by maximum processing times
within the function, it will be supplemented by
to meet timing requirements
- Describes communication between
Components, nodes, processes, etc.



DESCRIPTION OF GEOMETRY OF A SYSTEM

Concept drawing

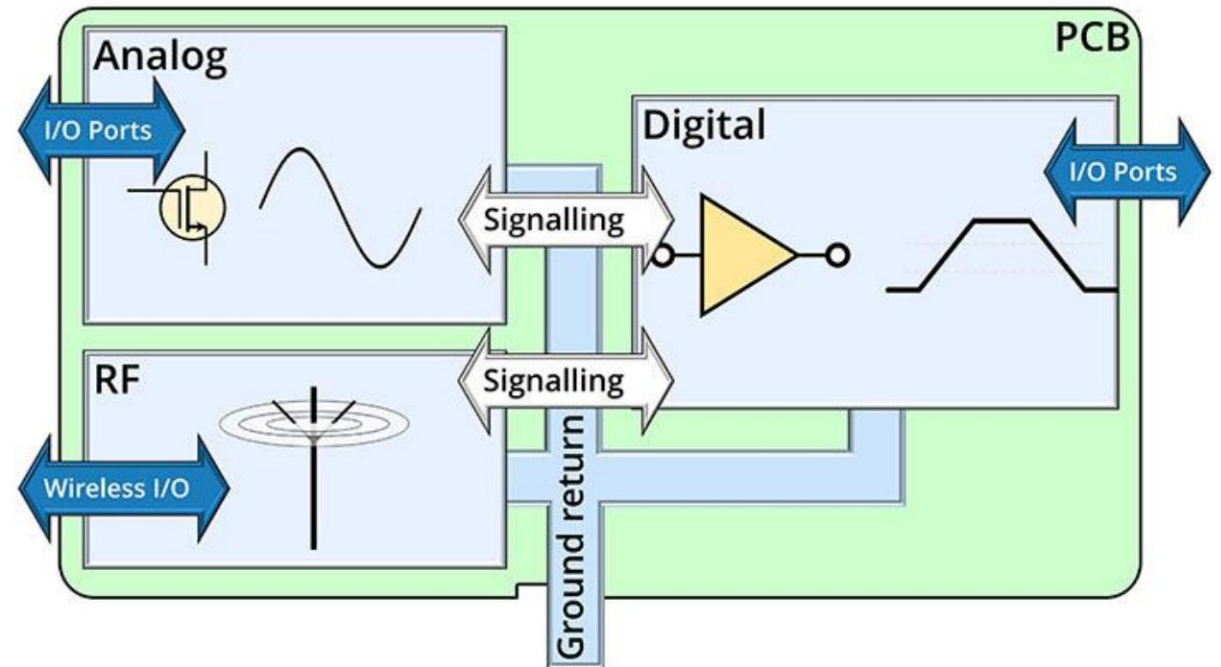
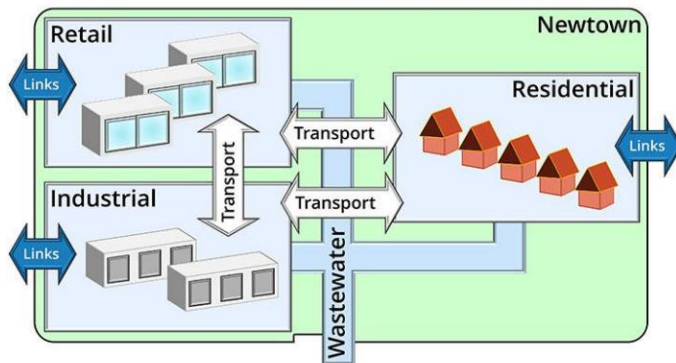
- enables extremely **fast visualization** of products
- relevant if innovative design is necessary
- useful for innovative products with multiple
Design cycles (e.g. in combination with spiral model or design thinking)
- Tools: Pen and paper



(Source: PCV Group)

Geometric description in 2D

- Defines the division of space of any kind objects
- Examples:
 - Floor plan of the factory hall
 - Room layout in the house

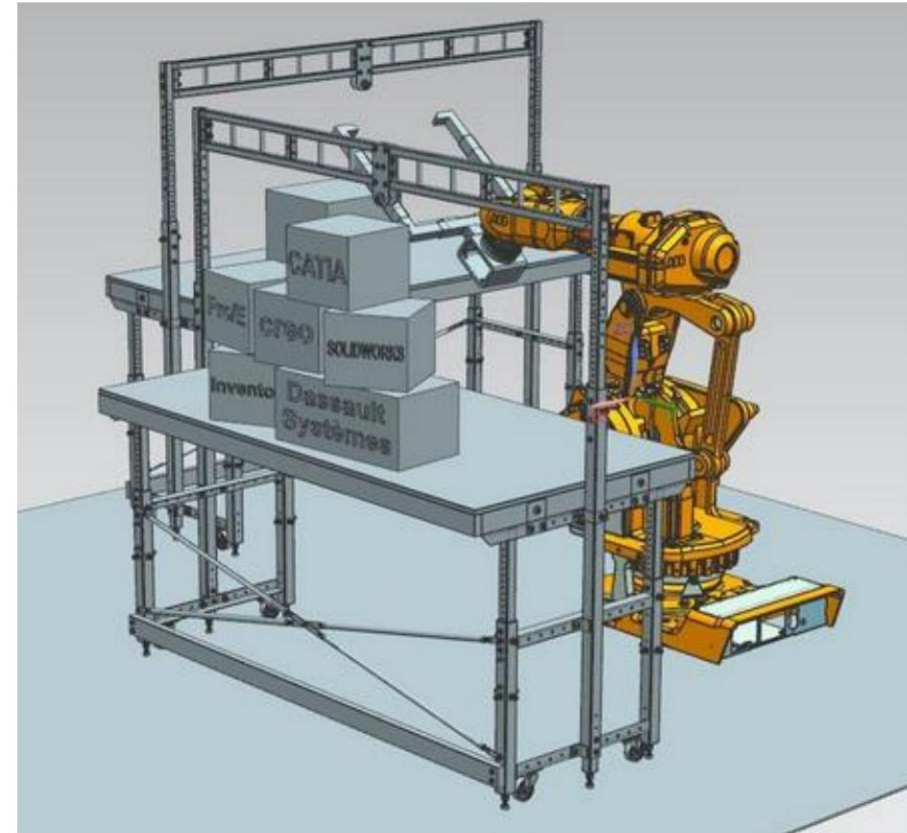


Planning the distribution of the assemblies on the printed circuit board

(Source: CADSTAR)

Geometric description in 3D

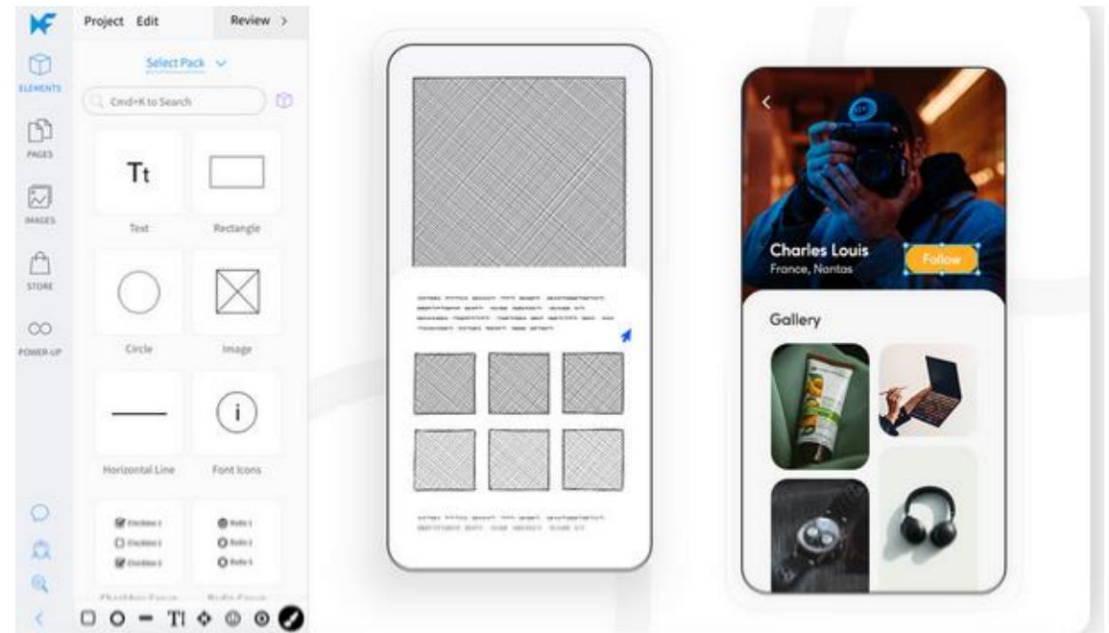
- **early spatial arrangement** using existing or simply 3D models
- Representation using simple geometries (sphere, plate, cylinder) possible
- enables the identification of problems (Narrow passages, missing supports)
- Tools: Fusion 360, Unity, Webots



(Quelle: grabcad)

Geometric division of the graphic interface

- Elements for the graphical user interface are displayed
- Tools: Figma, PlantUML



(Quelle: MockFlow)

INTERFACES- DESCRIPTION

Identify and specify interfaces

- An interface is part of a system/component and serves for communication
- Among other things, a distinction is made between:
 - **Data interface** – Data as interfaces between programs
 - **Machine interface** – Interfaces between physical systems
 - **Hardware interface** – Interfaces between physical systems in computer technology
 - **Network interface** – Interfaces between network components
 - **Software interface** – Interfaces between programs or software components
among each other
 - **User interface** (human-machine interface) – Interfaces between humans and devices



Identify and specify interfaces

- Interfaces are described, among other things, with
 - Circuit diagrams: used in bus systems to illustrate the wiring
 - Tables: Input/output signals, signal levels, message packets
 - Sequence diagrams: Representation of communication flow
 - SW component description: shows the interface between software components within a software application
- Describing interfaces is essential and requires abstraction skills.
- The author of the interface must read from the perspective of the target audience.

Specify software interfaces

- An **API (Application Programming Interface)** is a set of commands, functions, protocols and Objects that programmers can use to create software or interact with an external system.
- It provides developers with standard commands for performing common operations, so that Codes do not need to be rewritten from scratch.
- The API – also **called the programming interface** – enables applications to communicate with each other. The API is not the database or even the server, but rather the code that manages the access points for the server and facilitates communication.

(Source: talend.com)

Brainstorming zu APIs



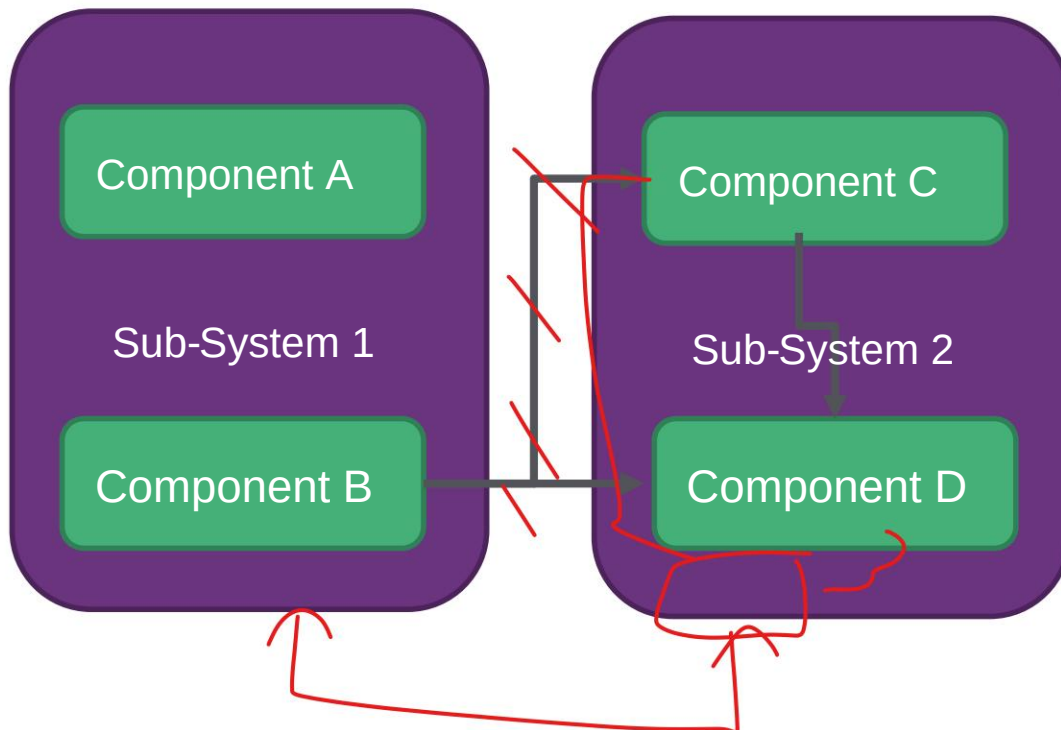
Which APIs are you familiar with?

Search the web for official documentation.

Coupling degree of components

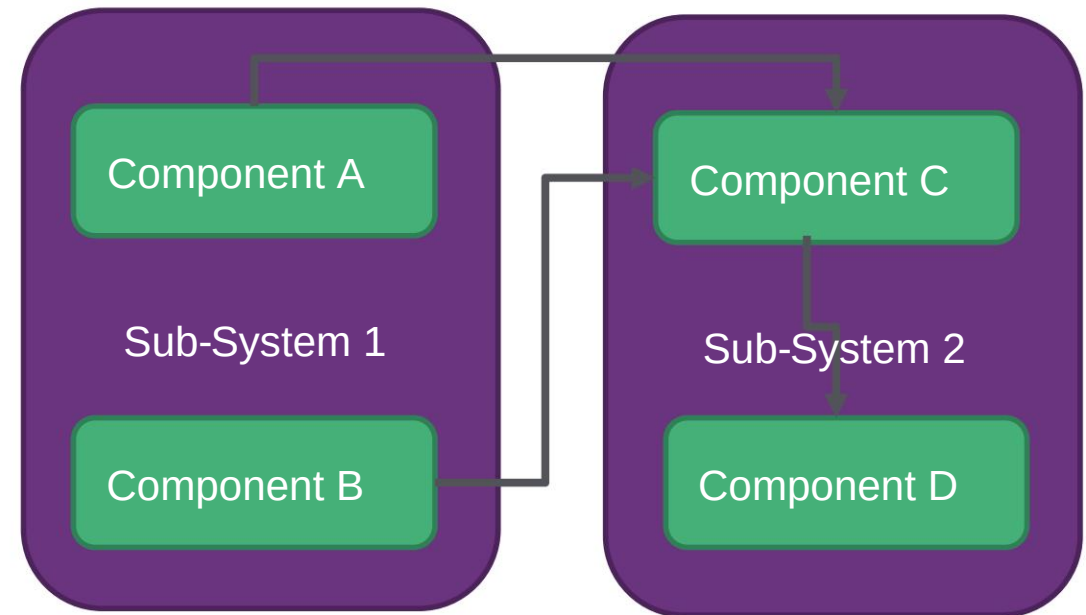
Low coupling

Easy maintenance
Low development effort



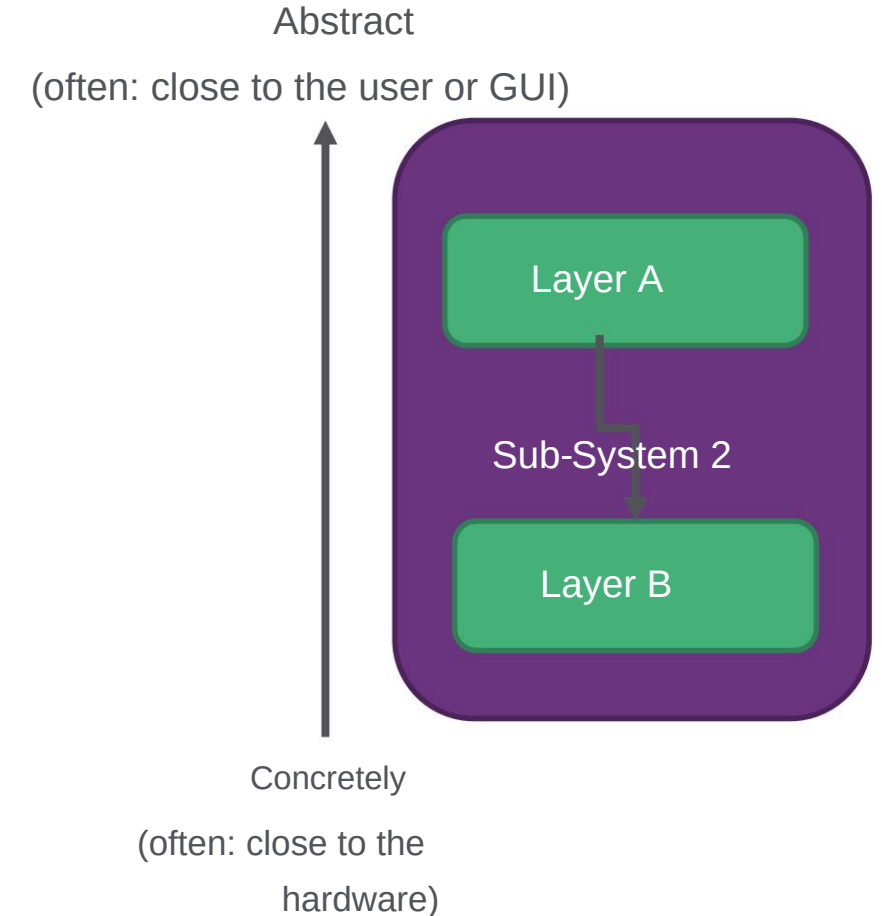
High coupling

Heavy maintenance,
more work in development

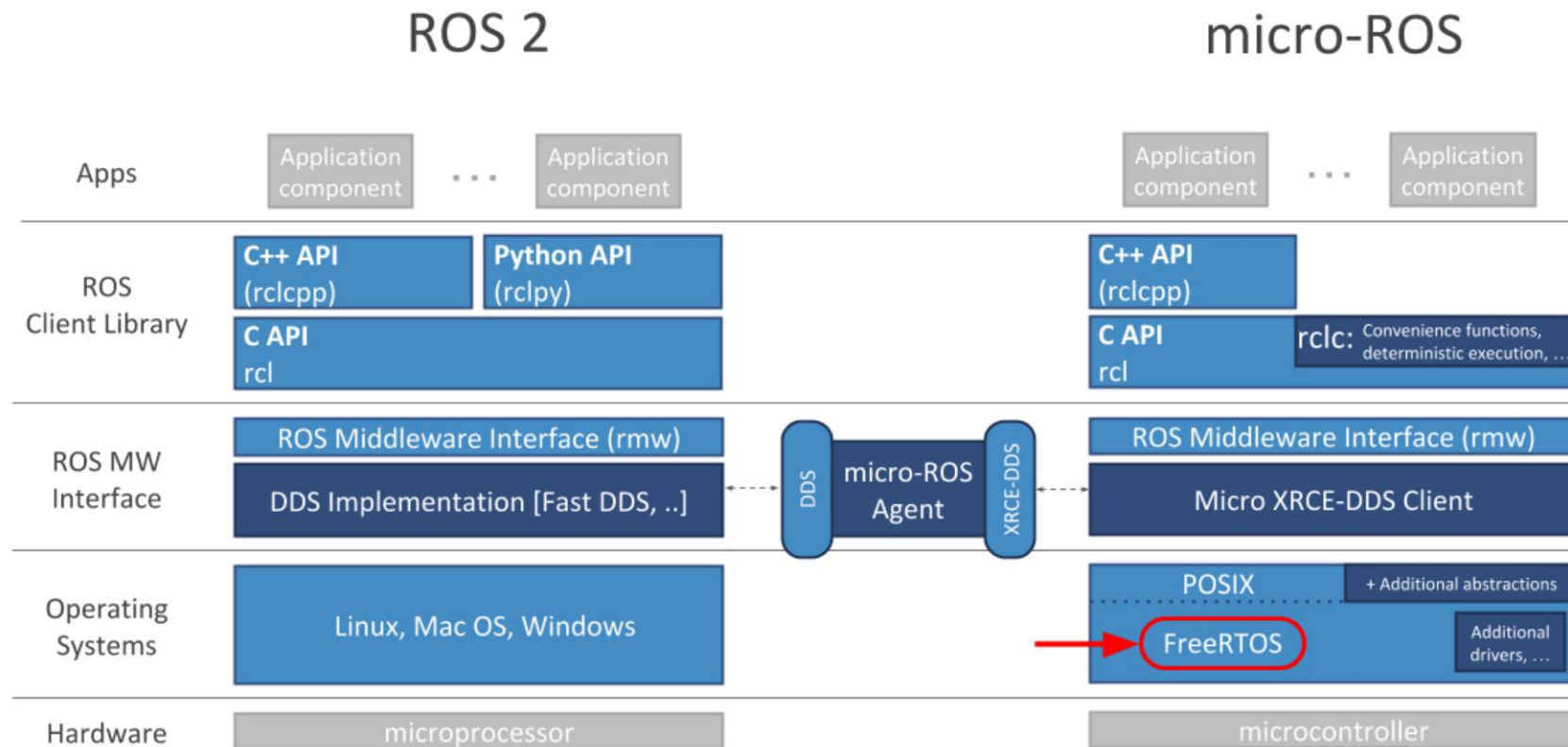


Representation as layers

- Each lower layer provides services for the layer(s) above it.
Shift(s) ready
- Examples:
 - OSI layer model for network protocols (7 layers)
 - Operating systems
- **Closed architecture:** a layer may only access the layer directly below it
(otherwise it is called an open architecture)



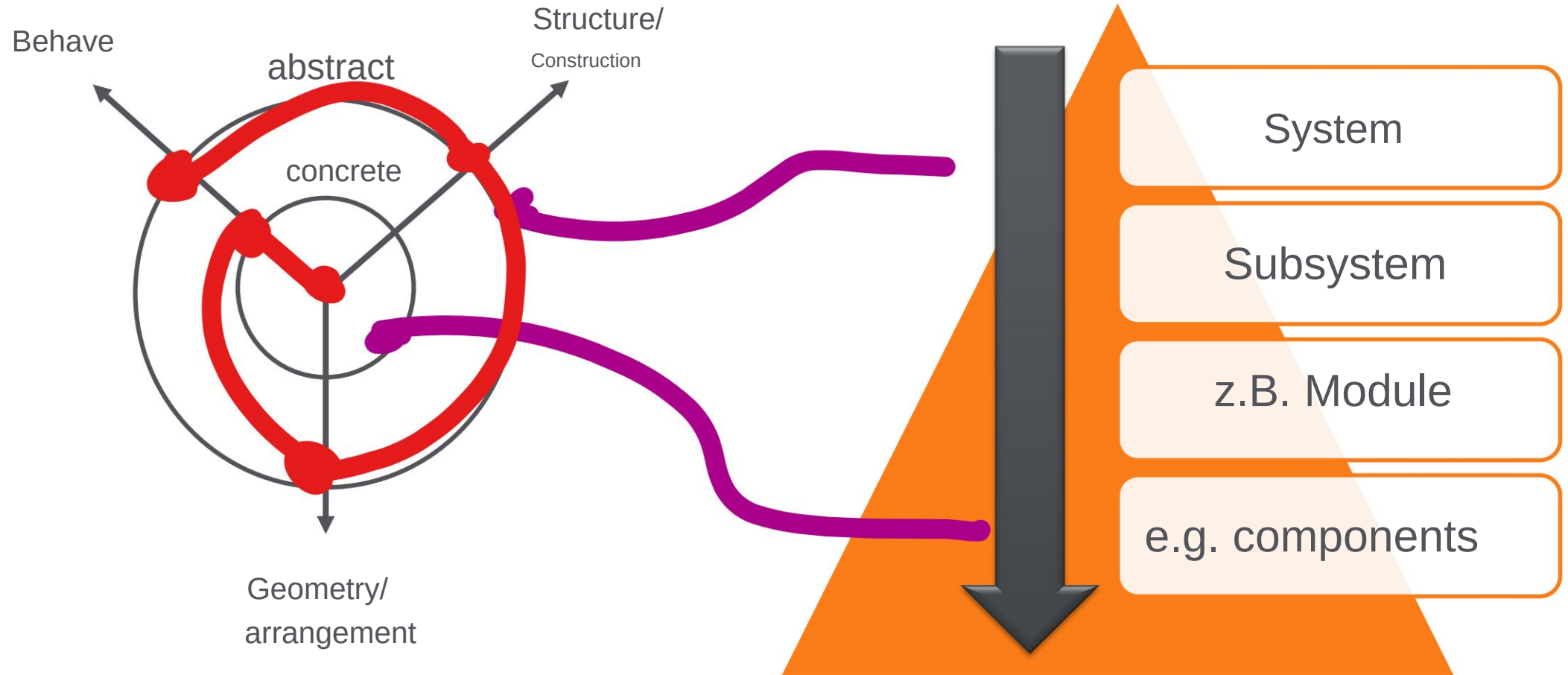
Example: Software stack of ROS and micro-ROS



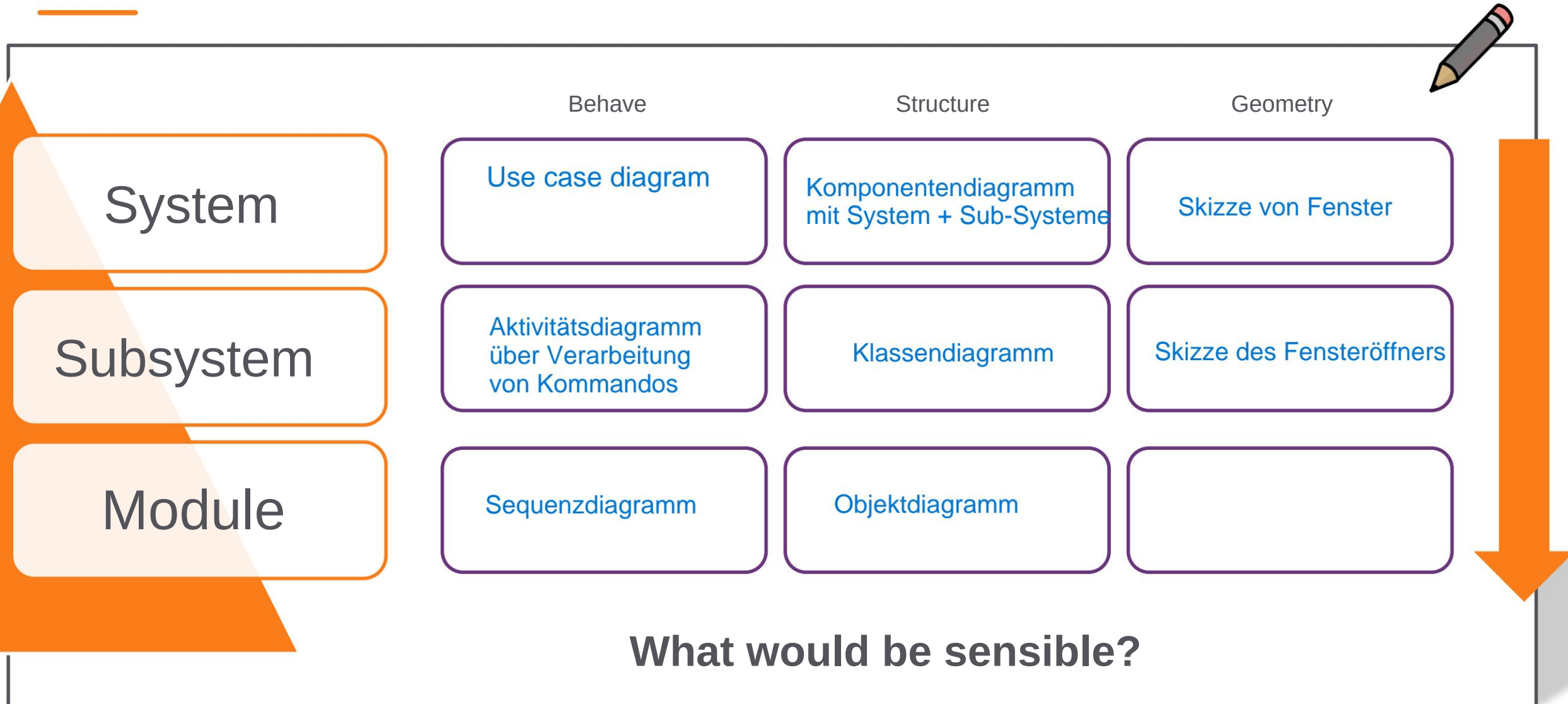
(Source: freertos.org)

SYSTEM ARCHITECTURE CREATE

Creating the system architecture top-down



Creating the system architecture top-down



Notice

- MECE approach: Mutually Exclusive Collectively Exhaustive means here:
 - ~~Komponenten~~
– Work packages **do not overlap** (= mutually exclusive)
 - ~~Komponenten~~
– Work packages **together form the overarching** work package or sub-project (collectively exhaustive)

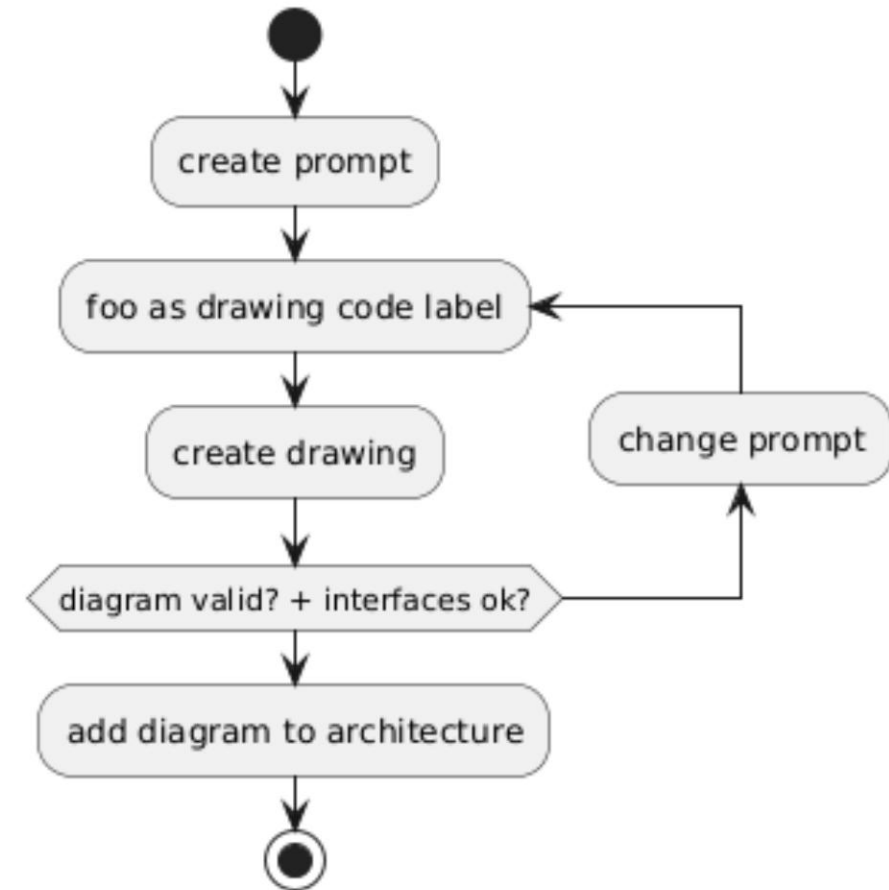
AI-BASED SYSTEM DESIGN

AI support in system design

- AI can help create the first draft and overcome the obstacles of the proverbial "blank page"
reduce
- Especially when there are few restrictive conditions, this can mark the beginning of the
Accelerate the design process
- PlantUML and Mermaid allow for easy modification of the charts, which is why adapting an initial one
Design is quite possible

Workflow

- The result must be checked for
 - Completeness
 - Interfaces
 - Plausibility of logic/designations
- Ideally, Prompt + Chatbot + Drawing Code
Save for future use
- Once several diagrams are already present
Extensive prompt engineering required



WORKSHOP

Exercise on system definition (Work in groups of 2 to 3, approx. 30 min.)



Objective: Design of a system for intelligent roller shutter control

To do this, create a system architecture manually **or** with AI support, using the following diagrams that describe the most important aspects. Only software components are considered.

Top Level: Component diagram, use case diagram, sequence diagram for communication between 2-3 of the components from the component diagram

Mid Level: Activity Diagram for Control Unit

At the end of the allotted time, place your diagrams on the digital whiteboard.

If you are working with AI support, mention Chatbot + Prompt in the whiteboard.